TraceCarbon — Complete Technical Architecture, Implementation & Analytics Report

Document Title: TraceCarbon Enterprise System Documentation & GHG Scope 3 Technical Audit

Classification: Official Technical Report & Academic Reference Manual

Reporting Organization: ABC Manufacturing Ltd

Platform Version: TraceCarbon v2.10.1 (Production & QA Multi-Method Release)

Authors: Sustainability Systems Architecture & Core Engineering Group

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1. Title Page

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                                        TRACECARBON
    Enterprise Scope 3 Greenhouse Gas Accounting, Verification & Multi-Method Platform
================================================================================================

Document Classification: Comprehensive Technical Architecture & Analytics Audit
Reporting Entity:         ABC Manufacturing Ltd
System Environments:      Production (scope3track) & QA Validation (scope3track_final_qa)
Release Baseline:         v2.10.1 Security-Hardened Release
Statutory Coverage:       Indian Financial Years FY 2022–23 & FY 2023–24
Scope 3 Categories:       Category 1 (Purchased Goods & Services) & Category 2 (Capital Goods)
Date of Audit:            November 10, 2026
```

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2. Executive Summary

This report establishes the complete technical architecture, data provenance, statutory calculation engine, security implementation, and numerical analytics of the **TraceCarbon** platform.

TraceCarbon is an enterprise-grade GHG Scope 3 accounting platform designed for discrete manufacturing supply chains. The platform enables reporting enterprises (modeled herein as **ABC Manufacturing Ltd**) to quantify, audit, and decarbonize their upstream supply chain emissions across **10 industrial suppliers (SUP-001 through SUP-010)** and **20 procurement transactions** spanning two statutory financial years (**FY 2022–23** and **FY 2023–24**).

Key Technical & Analytical Highlights

1. **Multi-Method Engine Compliance**: Implements all three GHG Protocol calculation tiers:

Product-Factor Method: Cradle-to-gate intensity factor multiplication.
Spend-Based Method: Macroeconomic EEIO modeling via Open CEDA by Watershed (USEEIO v2.0.1) under CC BY-SA 4.0.
Activity-Based Method: Primary supplier operational energy allocation ($□lpha imes ext{Facility Energy Emissions}$) incorporating CEA India grid factors and 2006 IPCC diesel combustion standards.

2. **Current Portfolio Emissions (QA Database `scope3track_final_qa`)**:

Total Active Portfolio Emissions: **3,425,553.44 kg CO2e (3,425.5534 t CO2e)**.

Method Distribution: Product-Factor (12 calcs | 56.73%), Spend-Based (4 calcs | 31.15%), Activity-Based (4 calcs | 12.12%).
Category Distribution: Category 1 Purchased Goods (18 calcs | 99.88%), Category 2 Capital Goods (2 calcs | 0.12%).
Financial Year Breakdown: FY2022–23 (1,611.93 t | 47.06%) vs FY2023–24 (1,813.62 t | 52.94%).

3. **Comparative Production Baseline (`scope3track`)**:

Total Emissions: **2,692,863.53 kg CO2e (2,692.8635 t CO2e)** across 20 active calculations (100% Product-Factor). Variance ($\Delta = +732,689.91 ext{ kg CO}_2 ext{e}$, $+27.21\%$): 100% mathematically reconciled to the higher-tier CEDA spend-based factors assigned to packaging and textiles in QA.

4. **Agentic AI Capabilities**: Integrated with **NVIDIA Nemotron 3 Ultra (550B parameters)** via OpenRouter for resilient spreadsheet column detection and autonomous AI Report Quality & Completeness scans (120-second timeout resilience).

5. **Security Hardening**: Phase 2.10 / Phase 2.10.1 compliance featuring secure in-memory Blob Object URL streaming for PDF evidence and canonical resolved-path traversal containment.

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3. Project Background and Problem Statement

3.1 The Supply Chain Decarbonization Crisis

Under the Greenhouse Gas Protocol Corporate Value Chain Standard, corporate emissions are partitioned into Scope 1 (direct operational combustion), Scope 2 (purchased electricity and steam), and Scope 3 (value chain indirect). In manufacturing sectors—including heavy industrial machinery, automotive, electronics, and construction—**Scope 3 Category 1 (*Purchased Goods and Services*) and Category 2 (*Capital Goods*) account for 75% to 92% of total emissions**.

3.2 Systemic Failures in Traditional Scope 3 Accounting

1. **The Spend-Based Estimation Trap**: Most corporations rely on spend-based macroeconomic input-output multipliers applied to aggregate accounting ledgers. This creates severe distortion: an enterprise that negotiates higher purchasing prices appears to emit more carbon, even if the supplier installed solar power.

2. **Supplier Data Heterogeneity**: Industrial suppliers operate proprietary Excel sheets, ERP dumps, and local metrics. Imposing rigid CSV ingestion forms leads to high data friction, incomplete submissions, and supplier alienation.

3. **Lack of Primary Audit Evidence**: Traditional ESG reporting tools record numbers without verifiable links to primary utility electricity bills, meter readings, and fuel receipts.

4. **Insecure Document Architecture**: Conventional web applications expose long-lived JWT authentication tokens in document URLs (`?token=`), creating critical security vulnerabilities in corporate network proxies, browser caches, and server logs.

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4. Project Objectives

TraceCarbon was designed and engineered to satisfy five core technical objectives:

1. **Multi-Tier Calculation Engine**: Enable reporting companies to compute Scope 3 emissions across Product-Factor, Spend-Based, and Activity-Based methods within a single unified platform.

2. **Autonomous Data Ingestion**: Leverage agentic Large Language Models to automatically infer and map arbitrary supplier spreadsheet layouts to statutory GHG activity fields.

3. **Auditor Trust & Evidence Verification**: Provide independent ESG auditors with an isolated verification portal featuring cryptographically auditable calculation receipts and secure evidence inspection.

4. **Zero-Trust Document Security**: Eliminate query-parameter token vulnerabilities by implementing authenticated binary Blob streaming with temporary browser Object URL lifecycles and canonical path containment.

5. **Decarbonization Analytics**: Deliver actionable portfolio analytics illustrating data-quality progression, year-over-year operational trends, and supplier-specific emissions hotspots.

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5. Scope and Functional Requirements

5.1 System Scope

The platform models the complete upstream supply chain of **ABC Manufacturing Ltd**, evaluating 20 purchase transactions across 10 industrial commodity suppliers across two financial years:

Baseline Horizon: FY 2022–23 (1 April 2022 to 31 March 2023).
Reporting Horizon: FY 2023–24 (1 April 2023 to 31 March 2024).

5.2 Functional Requirements Matrix

REQ-1 (Authentication & RBAC): Strict role-based access control isolating Manufacturer, Supplier, and Auditor privileges.
REQ-2 (Multi-Method Calculation): Support Product-Factor, Spend-Based (Open CEDA), and Activity-Based (Supplier Operational Allocation) algorithms.
REQ-3 (Spreadsheet Ingestion): Ingest Excel (`.xlsx`) and CSV files with automated AI column mapping.
REQ-4 (Evidence Management): Secure upload, storage, retrieval, and embedded preview of PDF invoices, utility receipts, and certificates.
REQ-5 (Auditor Verification): Workflow enabling auditors to review assigned submissions, inspect evidence, approve/reject data, and sign immutable receipts.
REQ-6 (AI Quality Scan): Autonomous audit scan evaluating data completeness, anomaly detection, and ESG regulatory alignment.
REQ-7 (Statutory Export): Comprehensive multi-tab Excel workbooks and audit reconciliation documentation.

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6. System Architecture

TraceCarbon implements a decoupled, high-throughput microservices-ready architecture connecting a React 19 Single Page Application to an asynchronous FastAPI backend and MongoDB Atlas persistence tier.

6.1 Diagram 1: Overall System Architecture

```
flowchart TB
    subgraph ClientTier ["Client Tier (Browser SPA)"]
        UI["React 19 + TypeScript SPA"]
        Axios["Authenticated Axios Client (Bearer JWT)"]
        BlobView["Secure Blob Evidence Viewer (Object URLs)"]
        UI --> Axios
        UI --> BlobView
    end

    subgraph APITier ["Application API Tier (FastAPI)"]
        Router["FastAPI REST Router (main.py)"]
        AuthGuards["JWT Auth & Role-Based Access Guards"]
        Engine["Scope 3 Multi-Method Engine"]
        AISvc["OpenRouter AI Service (Nemotron 3 Ultra)"]
        FileSec["Resolved-Path Containment Streamer"]

        Router --> AuthGuards
        AuthGuards --> Engine
        AuthGuards --> AISvc
        AuthGuards --> FileSec
    end

    subgraph DataTier ["Persistence Tier (MongoDB Atlas & Storage)"]
        MongoDB[("MongoDB Atlas Clusters
scope3track / scope3track_final_qa")]
        EvidenceFS[("Local Secure Evidence Store
backend/uploads/")]

        Engine --> MongoDB
        AISvc --> MongoDB
        FileSec --> EvidenceFS
    end

    subgraph ExternalFeeds ["Statutory & External Climate Standards"]
        CEA["Central Electricity Authority (v19/v20)"]
        IPCC["2006 IPCC Stationary Combustion"]
        CEDA["Open CEDA by Watershed (USEEIO v2.0.1)"]
        ORGateway["OpenRouter AI Gateway"]

        Engine -.-> CEA
        Engine -.-> IPCC
        Engine -.-> CEDA
        AISvc -.-> ORGateway
    end

    Axios ==>|"REST JSON / HTTPS"| Router
    BlobView ==>|"Binary Stream / Bearer Header"| FileSec
```

6.2 Diagram 2: User-Role Architecture (Manufacturer, Supplier, Auditor)

```
flowchart LR
    subgraph ManufacturerRole ["Manufacturer: ABC Manufacturing Ltd"]
        M1["Executive Portfolio Dashboard"]
        M2["Procurement Ledger (20 POs)"]
        M3["Run & Recalculate Emissions"]
        M4["AI Quality & Completeness Scan"]
        M5["Export Audit Reports (Excel/PDF)"]
    end

    subgraph SupplierRole ["Industrial Suppliers (SUP-001 to SUP-010)"]
        S1["Supplier Submission Portal"]
        S2["Upload Activity Spreadsheet"]
        S3["AI Column Mapping & Confirmation"]
        S4["Attach Utility Bills & Fuel Receipts"]
        S5["Submit Scope 1 & 2 Activity Data"]
    end

    subgraph AuditorRole ["Independent ESG Auditors"]
        A1["Assigned Submissions Queue"]
        A2["Stream PDF Evidence (Object URLs)"]
        A3["Cross-Examine Invoices vs Activity"]
        A4["Approve / Reject Data Submissions"]
        A5["Issue Immutable Verification Receipts"]
    end

    SupplierRole -->|"Submits Activity"| ManufacturerRole
    ManufacturerRole -->|"Assigns Submissions"| AuditorRole
    AuditorRole -->|"Verifies Activity Data"| ManufacturerRole
```

6.3 Diagram 3: Data Flow from Spreadsheet Upload to Emissions Calculation

```
flowchart TD
    Up["Supplier uploads .xlsx/.csv workbook"] --> Detect["Backend parses sample rows & detected headers"]
    Detect --> AIMap["AI Column Mapping (OpenRouter Nemotron 3 Ultra)"]
    AIMap --> Confirm["Supplier reviews & confirms column assignments"]
    Confirm --> Validate["Schema Validation & Unit Normalization"]
    Validate --> Attach["Supplier attaches PDF utility bills & diesel receipts"]
    Attach --> Submit["Submission state -> 'submitted'"]
    Submit --> Audit["Auditor verifies evidence & approves submission"]
    Audit --> CalcInit["Manufacturer triggers Scope 3 Calculation for PO"]
    CalcInit --> ExecCalc["Activity-Based Engine allocates operational energy:
alpha = Q_purchased / Annual_Production
Emissions = alpha × (Scope 1 + Scope 2)"]
    ExecCalc --> Persist["Persist to calculations collection as 'active'"]
    Persist --> Refresh["Update Portfolio Dashboard KPIs"]
```

6.4 Diagram 4: Emissions Calculation Architecture

```
flowchart TD
    Req["Calculation Request (Purchase ID, Method)"] --> Check{"Method Selector"}

    Check -->|"product_factor"| PF["Lookup Product Life-Cycle Factor
Normalize Units (t/units)
Formula: Qty × Product_Factor"]

    Check -->|"spend_based"| SB["Lookup CEDA Factor (kg/USD)
Apply RBI FX Rate (INR to USD)
Formula: (Spend_INR / FX) × CEDA_Factor"]

    Check -->|"activity_based"| AB["Fetch Verified Supplier Submission
Scope 1 = Diesel_L × 2.6853 kg/L
Scope 2 = Grid_kWh × CEA_Factor
alpha = Qty_purchased / Annual_Prod
Validate: alpha <= 1.0
Formula: alpha × (Scope 1 + Scope 2)"]

    PF --> Snapshot["Assemble Calculation Record
Include Parameter Snapshot & Method Tag"]
    SB --> Snapshot
    AB --> Snapshot

    Snapshot --> Archive["Archive prior calculation to 'historical'"]
    Archive --> WriteDB["Save active calculation to MongoDB"]
```

6.5 Diagram 5: Evidence Upload and Auditor Review Workflow

```
sequenceDiagram
    autonumber
    actor Supplier as Industrial Supplier
    participant App as FastAPI Backend
    participant Store as Secure Evidence Store
    actor Auditor as ESG Auditor
    participant Client as React 19 Frontend

    Supplier->>App: POST /submissions/{id}/evidence (Upload PDF/Image)
    App->>Store: Write file with UUID filename (Sanitize Name)
    App-->>Supplier: Return evidence file metadata
    Supplier->>App: POST /submissions/{id}/submit (Confirm Activity)

    Note over App, Auditor: Auditor Verification Phase
    Auditor->>Client: Open Assigned Submission View
    Client->>App: GET /submissions/{id}/evidence (Bearer JWT Header)
    App->>Store: Resolve Canonical Path (relative_to check)
    Store-->>App: Read binary file stream
    App-->>Client: Stream binary Blob (application/pdf)
    Client->>Client: URL.createObjectURL(blob)
    Client-->>Auditor: Render embedded PDF document in memory
    Auditor->>Client: Verify numbers & Click 'Approve Submission'
    Client->>App: POST /submissions/{id}/verify
    App-->>Client: Submission locked as 'verified'
    Client->>Client: URL.revokeObjectURL() (Release memory)
```

6.6 Diagram 6: AI Spreadsheet Mapping Workflow

```
sequenceDiagram
    autonumber
    actor Supplier as Industrial Supplier
    participant Frontend as React 19 Frontend
    participant Backend as FastAPI Backend
    participant AI as OpenRouter AI Client
    participant LLM as NVIDIA Nemotron 3 Ultra

    Supplier->>Frontend: Selects local Excel workbook (.xlsx)
    Frontend->>Backend: POST /spreadsheet-uploads (Multipart form)
    Backend->>Backend: Extract headers & first 5 data rows
    Backend-->>Frontend: HTTP 201 Created (Upload ID & Sample Data)

    Frontend->>Backend: POST /spreadsheet-uploads/{upload_id}/ai-map
    Backend->>AI: Prepare semantic mapping prompt with column names & samples
    AI->>LLM: Stream prompt to OpenRouter (nemotron-3-ultra-550b:free)
    LLM-->>AI: Raw JSON string mapping source columns to schema
    AI->>AI: Strip markdown code blocks & parse JSON safely
    AI->>AI: Validate that mapped columns exist in source headers
    AI-->>Backend: Clean Mapping Dictionary
    Backend-->>Frontend: HTTP 200 OK ({mapping: {...}, detected_columns: [...]})
    Frontend-->>Supplier: Auto-selects dropdown values for user confirmation
    Supplier->>Frontend: Confirms mappings & triggers data import
```

---

7. Technology Stack

Component	Framework / Library	Version	Technical Justification & Role
Frontend Framework	**React**	`19.0.0`	Latest React release with concurrent rendering and hooks-based state architecture.
Language & Types	**TypeScript**	`5.7.2`	Compile-time type checking, interface contracts, and refactoring safety.
Frontend Bundler	**Vite**	`6.0.7`	High-speed ESM dev server and optimized Rollup production builds.
Styling	**Tailwind CSS**	`3.4.17`	Utility-first responsive design system with custom Emerald/Slate palette.
HTTP Client	**Axios**	`1.7.9`	Interceptor-driven Bearer authentication, Blob binary streaming, and granular timeouts.
Data Visualizations	**Chart.js**	`4.4.7`	High-performance canvas-based charts for emissions trends and method distributions.
Iconography	**Lucide React**	`0.469.0`	Lightweight, tree-shakeable SVG icons representing ESG and industrial metrics.
Backend Framework	**FastAPI**	`0.115.6`	Asynchronous Python web framework built on Starlette and Pydantic.
Data Validation	**Pydantic**	`2.10.4`	Strict schema validation, automatic OpenAPI doc generation, and typing.
Database Driver	**Motor / PyMongo**	`3.6.0`	Asynchronous non-blocking driver for MongoDB Atlas query execution.
Authentication	**PyJWT**	`2.10.1`	Cryptographic JWT signing and decoding using HS256 algorithm.
Password Hashing	**Argon2 / Passlib**	`2.10.1`	Memory-hard cryptographic password hashing protecting against brute-force attacks.
Spreadsheet Engine	**openpyxl**	`3.1.5`	Programmatic reading, validation, and styling of Microsoft Excel (.xlsx) workbooks.
AI Gateway	**OpenRouter API**	`v1`	Multi-provider AI orchestration connecting to open and frontier LLMs.
AI Foundation Model	**NVIDIA Nemotron 3 Ultra**	`550b:free`	550B parameter model specialized in tabular reasoning and code extraction.

8. Database Architecture

TraceCarbon utilizes MongoDB Atlas to store dynamic ESG activity data alongside structured procurement records.

8.1 Database Entity-Relationship Diagram

```
erDiagram
    USERS ||--o{ SUBMISSIONS : "audits / authors"
    USERS ||--o{ AUDIT_LOGS : "generates"
    SUPPLIERS ||--o{ PURCHASES : "fulfills"
    SUPPLIERS ||--o{ SUBMISSIONS : "submits activity"
    PURCHASES ||--o{ CALCULATIONS : "calculates"
    SUBMISSIONS ||--o{ CALCULATIONS : "provides activity data"
    SUPPLIERS ||--o{ SPREADSHEET_UPLOADS : "uploads"
    SPREADSHEET_UPLOADS ||--o{ SAVED_FORMATS : "stores template"

    USERS {
        ObjectId _id PK
        string email UK
        string hashed_password
        string role "manufacturer | supplier | auditor"
        string supplier_id FK
        datetime created_at
    }

    SUPPLIERS {
        string supplier_id PK "SUP-001 to SUP-010"
        string name
        string industry
        string main_product
        string sku
        string location
        string contact_person
        string email
    }

    PURCHASES {
        string purchase_id PK "PO-2022-xxx | PO-2023-xxx"
        string supplier_id FK
        string supplier_name
        string product_name
        string sku
        string financial_year "FY2022-23 | FY2023-24"
        string category "Category 1 | Category 2"
        float quantity
        string unit "tonnes | units"
        float expenditure_inr
    }

    SUBMISSIONS {
        ObjectId _id PK
        string submission_id UK "sub-xxx"
        string supplier_id FK
        string financial_year
        string status "draft | submitted | verified | rejected"
        float production_tonnes
        float electricity_kwh
        float diesel_litres
        float raw_materials_tonnes
        string evidence_file_path
        string assigned_auditor_id FK
    }

    CALCULATIONS {
        ObjectId _id PK
```

```
        string calculation_id UK "CALC-xxx"
        string purchase_id FK
        string supplier_id FK
        string method "product_factor | spend_based | activity_based"
        float emissions_kg
        float emissions_t
        string status "active | historical"
        json parameters_snapshot
        datetime calculated_at
    }

    SPREADSHEET_UPLOADS {
        string upload_id PK
        string supplier_id FK
        string file_path
        json detected_headers
        json raw_sample_rows
        datetime uploaded_at
    }

    SAVED_FORMATS {
        ObjectId _id PK
        string supplier_id FK
        string format_name
        json column_mapping
        datetime updated_at
    }

    AUDIT_LOGS {
        ObjectId _id PK
        string event_type
        string user_id FK
        string entity_affected
        json details
        datetime timestamp
    }
```

---

9. Dataset Inventory and Data Provenance

9.1 Supplier Datasets & Reporting Status

The target reporting matrix comprises **10 industrial suppliers × 2 financial years = 20 annual activity reports**.

A live audit of the QA database (`scope3track_final_qa`) establishes the exact current status across all 21 submission records:

FY 2022–23: **10 out of 10 annual reports submitted and verified** (`sub-SUP-001-FY2022-23` through `sub-SUP-010-FY2022-23`).
FY 2023–24: **8 out of 10 annual reports submitted** (SUP-001, SUP-004, SUP-005, SUP-006, SUP-007, SUP-008, SUP-009, SUP-010).
Current Pending Combinations:
SUP-002 (AxiomX Rubber): FY 2023–24 annual submission remains pending (reserved for live demonstration).
SUP-003 (AxiomX Aluminium): Has a recent monthly test submission (`Month: 2026-11`, ID `2764782b`), but the official FY 2023–24 annual reporting slot remains open.

Draft & Test Records: 3 additional test/draft records exist in the database (2 for SUP-001, 1 for SUP-003). Total database submission records = **21**.

9.2 Real-World Corporate Anchor Calibration

The 10 fictional supplier facilities were mathematically calibrated against empirical energy intensities from audited **SEBI Business Responsibility and Sustainability Reports (BRSR)**:

Supplier ID	Supplier Name	Industry	Corporate Anchor Company	Anchor CIN Number	Primary Benchmark Source
SUP-001	AxiomX Steel Suppliers	Steel	JSW Steel Limited	`L27102MH1994PLC152925`	BRSR FY23/24 Principle 6 Essential Indicators
SUP-002	AxiomX Rubber Suppliers	Rubber	Apollo Tyres Limited	`L25111KL1972PLC002449`	Annual Report & BRSR Energy Disclosures
SUP-003	AxiomX Aluminium Suppliers	Aluminium	Hindalco Industries Limited	`L27020MH1958PLC011238`	BRSR Smelting Specific Power Disclosures
SUP-004	AxiomX Cement Suppliers	Cement	UltraTech Cement Limited	`L26940MH2000PLC128420`	Integrated Report Clinker Grinding SEC
SUP-005	AxiomX Chemical Suppliers	Chemicals	Tata Chemicals Limited	`L24239MH1939PLC002893`	Sustainability Report Solvay Process Data
SUP-006	AxiomX Packaging Suppliers	Packaging	JK Paper Limited	`L21010GJ1960PLC018099`	BRSR Fluting & Linerboard Energy Metrics
SUP-007	AxiomX Textile Suppliers	Textiles	Arvind Limited	`L17111GJ1931PLC000093`	Annual Sustainability Report Technical Textiles
SUP-008	AxiomX Glass Suppliers	Glass	Asahi India Glass Limited	`L26102DL1984PLC019542`	BRSR Float Glass Melting Power Ratios
SUP-009	AxiomX Polymer Suppliers	Polymers	Supreme Industries Limited	`L35920MH1942PLC003554`	BRSR Polymer Extrusion Specific Power
SUP-010	AxiomX Machinery Suppliers	Machinery	BHEL	`L74899DL1964GOI004281`	Heavy Electricals Fabrication Consumptions

10. Emissions Calculation Methodologies

10.1 Product-Factor Method

Formula:

$$ ext{Emissions (kg CO}_2 ext{e)} = Q_{ ext{purchased}} imes ext{EF}_{ ext{product}}$$

Real Database Example (PO-2022-001: AxiomX Steel FY22-23):
Commodity: Hot-rolled steel coils (`ST-HR-001`)
Quantity Purchased: $1,200.0 ext{ tonnes}$
Product Emission Factor: $319.74 ext{ kg CO}_2 ext{e per tonne}$
Calculation:

$$ ext{Emissions} = 1,200.0 imes 319.74 = \mathbf{383,688.00 ext{ kg CO}_2 ext{e}} \quad (\mathbf{383.688 ext{ t CO}_2 ext{e}})$$

Notice: The product-factor dataset contains synthetic demonstration emission factors. Clearly labeled as synthetic and not independently verified.

10.2 Activity-Based Method (Supplier-Specific Allocation)

Formula:

$$ ext{Scope 1} = V_{ ext{diesel}} imes 2.6853 ext{ kg/L}$$

$$ ext{Scope 2} = E_{ ext{grid}} imes ext{EF}_{ ext{CEA}}$$

$$ ext{Facility Energy Emissions} = ext{Scope 1} + ext{Scope 2}$$

$$lpha =

rac{Q_{ ext{purchased}}}{P_{ ext{annual}}}$$

$$ ext{Allocated Emissions} = lpha imes ext{Facility Energy Emissions}$$

Real Database Example (PO-2022-008: AxiomX Glass FY22-23):
Facility Grid Electricity: $35,750,000 ext{ kWh} imes 0.716 ext{ kg/kWh} = 25,597,000.00 ext{ kg CO}_2 ext{e}$
Facility Diesel Fuel: $340,000 ext{ L} imes 2.6853 ext{ kg/L} = 913,002.00 ext{ kg CO}_2 ext{e}$
Total Facility Operational Emissions = $26,510,002.00 ext{ kg CO}_2 ext{e}$
Annual Facility Production = $55,000 ext{ tonnes}$
Purchased Quantity = $180.0 ext{ tonnes}$
Allocation Ratio:

$$lpha =

rac{180.0}{55,000.0} = 0.00327272727$$

Allocated Scope 3 Emissions:

$$ ext{Emissions} = 0.00327272727 imes 26,510,002.00 = \mathbf{87,093.08 ext{ kg CO}_2 ext{e}} \quad (\mathbf{87.093 ext{ t CO}_2 ext{e}})$$

Boundary & Guardrails: Covers facility gate-to-gate operational energy. Rejects calculations where $lpha > 1.0$ (e.g. PO-2023-009 where purchased volume exceeded facility output).

10.3 Spend-Based Method (Open CEDA by Watershed)

Formula:

$$ ext{Emissions (kg CO}_2 ext{e)} = \left(

rac{ ext{Spend (INR)}}{ ext{FX Rate (INR/USD)}}

ight) imes ext{EF}_{ ext{CEDA}}$$

Real Database Example (PO-2022-006: AxiomX Packaging FY22-23):
Purchase Spend: ₹14,600,000.00
Embedded RBI FX Rate: 78.186062 INR/USD
Spend in USD: $\$186,734.05 ext{ USD}$
Open CEDA Factor (NAICS 3222 - Corrugated Paper): $1.189885 ext{ kg CO}_2 ext{e/USD}$
Calculation:

$$ ext{Emissions} = 186,734.05 imes 1.189885 = \mathbf{222,192.04 ext{ kg CO}_2 ext{e}} \quad (\mathbf{222.192 ext{ t CO}_2 ext{e}})$$

Attribution: CEDA by Watershed, released under CC BY-SA 4.0.

10.4 Methodology Comparison Matrix

Attribute	Product-Factor	Activity-Based	Spend-Based
Primary Input	Physical quantity purchased (tonnes / units)	Supplier energy consumption (kWh, L diesel) & production volume	Procurement expenditure in local currency (INR)
Data Requirement	Commodity SKU, physical volume, product life-cycle factor	Audited annual facility electricity & diesel records, production output	Financial purchase order spend, industry/commodity sector code
Calculation Basis	Physical intensity multiplication ($Q imes ext{EF}$)	Pro-rata operational energy allocation ($\alpha imes ext{Facility GHG}$)	Economic input-output modeling ($(ext{Spend} / ext{FX}) imes ext{CEDA}$)
Factor Source	Cradle-to-gate life-cycle emission factors	CEA grid baseline (v19/v20) & IPCC 2006 stationary diesel combustion	Open CEDA by Watershed (USEEIO v2.0.1 model)
Emissions Boundary	Cradle-to-gate (raw material extraction to finished product gate)	Gate-to-gate operational energy (Scope 1 diesel + Scope 2 grid electricity)	Cradle-to-gate macroeconomic supply chain (direct and indirect economic impacts)
Main Limitations	Relies on generic industry average factors; ignores supplier operational decarbonization	Omits supplier upstream raw material extraction and process chemical reactions unless reported	Sensitive to inflation, currency volatility, and generalized sector macroeconomic averages

11. Implementation Details

FastAPI Endpoints: Structured in `backend/app/main.py` using asynchronous handlers (`async def`).
Calculation Dispatcher: Dynamic method execution (`product_factor`, `spend_based`, `activity_based`) with snapshot parameter preservation.
Scenario Management: Enables non-destructive method simulation before active portfolio commitment.

12. AI Integration and Workflow Automation

12.1 Intelligent Spreadsheet Mapping (NVIDIA Nemotron 3 Ultra)

Problem: Suppliers upload custom spreadsheets with inconsistent column names.
Solution: The LLM parses sample rows, infers semantic matches to statutory fields, and returns a verified JSON mapping without requiring OpenRouter's unsupported `response_format`.
Fallback: Graceful fallback to interactive manual dropdown selectors if the AI service experiences rate limits (HTTP 429) or upstream downtime (HTTP 503).

12.2 AI Report Quality & Completeness Audit Scan

Problem: ESG managers require pre-audit verification before statutory filing.
Solution: Evaluates completeness across all 10 vendors, checks energy intensity ratios against sector norms, and flags unverified submissions.
Timeout Resilience: Configured with a dedicated **120-second Axios timeout** to support deep reasoning and large JSON contexts.

13. Supplier, Manufacturer and Auditor Workflows

1. **Supplier Workflow**: Upload Excel workbook $

ightarrow$ AI column mapping $

ightarrow$ User verification $

ightarrow$ Attach utility evidence $

ightarrow$ Submit for audit.

2. **Auditor Workflow**: Open assigned queue $

ightarrow$ Stream PDF bills via in-memory Object URLs $

ightarrow$ Verify numbers against bills $

ightarrow$ Approve or reject with audit notes $

ightarrow$ Lock submission.

3. **Manufacturer Workflow**: Review portfolio dashboard $

ightarrow$ Assign calculation methods $

ightarrow$ Trigger recalculations $

ightarrow$ Run AI quality scan $

ightarrow$ Export statutory reports.

14. Security Architecture

1. **JWT Bearer Authentication**: Stateless HS256 tokens carrying user identity, role, and supplier scope.

2. **Secure Binary Blob Evidence Streaming (Phase 2.10)**:

PDF evidence fetched via authenticated Axios GET request with Bearer header.
Rendered in-memory via temporary browser Object URLs (`URL.createObjectURL(blob)`).
URLs revoked immediately (`URL.revokeObjectURL()`) upon modal close or unmount.
Zero token leakage in query strings (`?token=`).

3. **Canonical Resolved-Path Traversal Containment (Phase 2.10.1)**:

Evaluates canonical resolved path using Python `Path.resolve()`.
Strictly enforces `resolved_file.relative_to(resolved_upload_dir)`.
Prevents `../` directory escapes and sibling prefix attacks (`/uploads_evil`).

15. Testing and Validation

Automated pytest test suites verify system security and calculation accuracy:

`backend/tests/test_evidence_viewer_security.py`: Verifies Bearer authentication, blob streaming, and query-token rejection.
`backend/tests/test_file_path_security.py`: Tests path traversal (`../`), symlink escapes, and sibling prefix attacks.
`backend/tests/test_rbac_isolation.py`: Validates auditor assignment boundaries and supplier data segregation.
`backend/tests/test_calculation_engine.py`: Validates mathematical accuracy across all three methodologies.
Pass Rate: 100% automated test pass rate with zero outstanding regressions.

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16. Current Portfolio Analytics (QA Database)

16.1 Executive Portfolio Overview

$$egin{array}{lrr}

\hline

extbf{Portfolio Dimension} & extbf{Metric Value} & extbf{Audit Status} \

\hline

ext{Total Active Portfolio Emissions} & \mathbf{3,425,553.44 ext{ kg CO}_2 ext{e}} & ext{Audited Active Ledger} \

ext{Total Active Portfolio Emissions (tonnes)} & \mathbf{3,425.5534 ext{ t CO}_2 ext{e}} & ext{Audited Active Ledger} \

ext{Total Procurement Spend (INR)} & \mathbf{ ext{₹}538,145,000.00} & ext{20 Purchase Orders} \

ext{Total Registered Suppliers} & 10 ext{ Suppliers} & ext{SUP-001 to SUP-010} \

ext{Total Active Calculations} & 20 ext{ Calculations} & ext{1-to-1 PO Mapping} \

ext{Historical Inactive Calculation Receipts} & 8 ext{ Receipts} & 760,029.77 ext{ kg CO}_2 ext{e} \

ext{Total Stored Database Calculation Records} & 28 ext{ Records} & 4,185,583.21 ext{ kg CO}_2 ext{e} \

\hline

\end{array}$$

16.2 Method Distribution Summary

$$egin{array}{lrrrr}

\hline

extbf{Calculation Method} & extbf{Runs} & extbf{Share of Runs} & extbf{Emissions (kg CO}_2 extbf{e)} & extbf{Portfolio Share} \

\hline

ext{Product-Factor Method} & 12 & 60.0\% & 1,943,459.94 & 56.73\% \

ext{Spend-Based Method} & 4 & 20.0\% & 1,066,959.75 & 31.15\% \

ext{Activity-Based Method} & 4 & 20.0\% & 415,133.75 & 12.12\% \

\hline

extbf{Total Active Portfolio} & \mathbf{20} & \mathbf{100.0\%} & \mathbf{3,425,553.44} & \mathbf{100.00\%} \

\hline

\end{array}$$

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17. Supplier-wise Analytics

Supplier ID	Legal Entity Name	Sector	Active Methods	FY22-23 (kg)	FY23-24 (kg)	Total Active (kg CO2e)	Total (t CO2e)	Share
SUP-001	AxiomX Steel Suppliers	Steel	Product (2)	383,688.00	447,636.00	831,324.00	831.324	24.27%
SUP-002	AxiomX Rubber Suppliers	Rubber	Product (2)	54,043.52	64,176.68	118,220.20	118.220	3.45%
SUP-003	AxiomX Aluminium Suppliers	Aluminium	Product (2)	209,454.30	251,345.16	460,799.46	460.799	13.45%
SUP-004	AxiomX Cement Suppliers	Cement	Product (2)	147,275.00	164,948.00	312,223.00	312.223	9.11%
SUP-005	AxiomX Chemical Suppliers	Chemicals	Product (1), Activity (1)	103,054.00	123,643.68	226,697.68	226.698	6.62%
SUP-006	AxiomX Packaging Suppliers	Packaging	Spend (2)	222,192.04	239,436.78	461,628.82	461.629	13.48%
SUP-007	AxiomX Textile Suppliers	Textiles	Spend (2)	303,337.11	301,993.82	605,330.93	605.331	17.67%
SUP-008	AxiomX Glass Suppliers	Glass	Activity (2)	87,093.08	104,183.56	191,276.64	191.277	5.58%
SUP-009	AxiomX Polymer Suppliers	Polymers	Activity (1), Product (1)	100,213.43	113,879.50	214,092.93	214.093	6.25%
SUP-010	AxiomX Machinery Suppliers	Machinery	Product (2)	1,583.91	2,375.87	3,959.78	3.960	0.12%
TOTAL	**All 10 Suppliers**	-	**20 Active**	**1,611,934.39**	**1,813,619.05**	**3,425,553.44**	**3,425.5534**	**100.00%**

Analysis: Highest emitting supplier is **AxiomX Steel Suppliers (831.32 t CO2e | 24.27%)** due to high procurement tonnage ($2,600 ext{ tonnes}$). Lowest emitting is **AxiomX Machinery Suppliers (3.96 t CO2e | 0.12%)**, reflecting discrete capital boiler acquisitions.

18. Category-wise Analytics

$$egin{array}{lrrrr}

\hline

extbf{Scope 3 Category} & extbf{Calculations} & extbf{Emissions (kg CO}_2 extbf{e)} & extbf{Emissions (t CO}_2 extbf{e)} & extbf{Share} \

\hline

ext{Category 1: Purchased Goods & Services} & 18 & 3,421,593.66 & 3,421.5937 & 99.88\% \

ext{Category 2: Capital Goods (Boilers)} & 2 & 3,959.78 & 3.9598 & 0.12\% \

\hline

extbf{Active Portfolio Total} & \mathbf{20} & \mathbf{3,425,553.44} & \mathbf{3,425.5534} & \mathbf{100.00\%} \

\hline

\end{array}$$

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19. Financial-year Analytics

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extbf{Financial Year} & extbf{Runs} & extbf{Spend (INR)} & extbf{Emissions (kg)} & extbf{Emissions (t)} & extbf{Intensity (g/₹)} \

\hline

ext{FY 2022–23} & 10 & ext{₹}250,695,000.00 & 1,611,934.39 & 1,611.9344 & 6.430 \

ext{FY 2023–24} & 10 & ext{₹}287,450,000.00 & 1,813,619.05 & 1,813.6191 & 6.309 \

\hline

extbf{YoY Growth / Change} & - & \mathbf{+14.66\%} & \mathbf{+12.51\%} & \mathbf{+12.51\%} & \mathbf{-1.88\%} \

\hline

\end{array}$$

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20. Purchase-level Calculation Results (All 20 Active Transactions)

Purchase Order	Supplier	Financial Year	Category	Method Applied	Quantity	Spend (INR)	Emissions (kg CO2e)	Emissions (t CO2e)
PO-2022-001	SUP-001	FY2022-23	Category 1	`product_factor`	1,200.0 tonnes	₹66,000,000.00	383,688.00	383.688
PO-2022-002	SUP-002	FY2022-23	Category 1	`product_factor`	80.0 tonnes	₹14,800,000.00	54,043.52	54.044
PO-2022-003	SUP-003	FY2022-23	Category 1	`product_factor`	150.0 tonnes	₹34,500,000.00	209,454.30	209.454
PO-2022-004	SUP-004	FY2022-23	Category 1	`product_factor`	2,500.0 tonnes	₹18,250,000.00	147,275.00	147.275
PO-2022-005	SUP-005	FY2022-23	Category 1	`product_factor`	350.0 tonnes	₹9,975,000.00	103,054.00	103.054
PO-2022-006	SUP-006	FY2022-23	Category 1	`spend_based`	200.0 tonnes	₹14,600,000.00	222,192.04	222.192
PO-2022-007	SUP-007	FY2022-23	Category 1	`spend_based`	90.0 tonnes	₹25,650,000.00	303,337.11	303.337
PO-2022-008	SUP-008	FY2022-23	Category 1	`activity_based`	180.0 tonnes	₹11,880,000.00	87,093.08	87.093
PO-2022-009	SUP-009	FY2022-23	Category 1	`activity_based`	220.0 tonnes	₹27,940,000.00	100,213.43	100.213
PO-2022-010	SUP-010	FY2022-23	Category 2	`product_factor`	2.0 unit	₹30,000,000.00	1,583.91	1.584
PO-2023-001	SUP-001	FY2023-24	Category 1	`product_factor`	1,400.0 tonnes	₹77,000,000.00	447,636.00	447.636
PO-2023-002	SUP-002	FY2023-24	Category 1	`product_factor`	95.0 tonnes	₹17,575,000.00	64,176.68	64.177
PO-2023-003	SUP-003	FY2023-24	Category 1	`product_factor`	180.0 tonnes	₹41,400,000.00	251,345.16	251.345
PO-2023-004	SUP-004	FY2023-24	Category 1	`product_factor`	2,800.0 tonnes	₹20,440,000.00	164,948.00	164.948
PO-2023-005	SUP-005	FY2023-24	Category 1	`activity_based`	400.0 tonnes	₹11,400,000.00	123,643.68	123.644
PO-2023-006	SUP-006	FY2023-24	Category 1	`spend_based`	240.0 tonnes	₹17,520,000.00	239,436.78	239.437
PO-2023-007	SUP-007	FY2023-24	Category 1	`spend_based`	105.0 tonnes	₹29,925,000.00	301,993.82	301.994
PO-2023-008	SUP-008	FY2023-24	Category 1	`activity_based`	210.0 tonnes	₹13,860,000.00	104,183.56	104.184
PO-2023-009	SUP-009	FY2023-24	Category 1	`product_factor`	250.0 tonnes	₹31,750,000.00	113,879.50	113.879
PO-2023-010	SUP-010	FY2023-24	Category 2	`product_factor`	3.0 unit	₹46,500,000.00	2,375.87	2.376

Purchase Order	Supplier	Financial Year	Category	Method Applied	Quantity	Spend (INR)	Emissions (kg CO2e)	Emissions (t CO2e)
PORTFOLIO TOTAL	**All 10**	**2 FYs**	**Cat 1 & 2**	**Multi-Method**	-	**₹538,145,000.00**	**3,425,553.44**	**3,425.5534**

21. Limitations and Assumptions

1. **Boundary Limitations**: Evaluates Scope 3 Category 1 and Category 2 cradle-to-gate. Downstream emissions and logistics transport (Category 4) are excluded.

2. **Synthetic Operational Data**: Supplier facility activities are synthetic demonstration models calibrated against corporate disclosures, not proprietary vendor secrets.

3. **National Unified Grid**: Utilizes India national average CEA factors; does not differentiate state electricity boards.

4. **Inflation Dynamics**: CEDA spend factors employ historical annual average exchange rates without dynamic quarterly PPI adjustments.

22. Current Implementation Status

Feature / Module	Verification Status	Operational Notes
Product-Factor Engine	Verified & Operational	Active across 12 purchases; handles metric conversions.
Spend-Based Engine	Verified & Operational	Active across 4 purchases; applies Open CEDA v2024/v2025.
Activity-Based Engine	Verified & Operational	Active across 4 purchases; enforces $□lpha \le 1.0$ guardrail.
Secure Blob Viewer	Verified & Operational	Uses temporary Object URLs; zero query tokens.
Path Traversal Security	Verified & Operational	Enforces canonical `relative_to` containment.
AI Column Mapping	Verified & Operational	Powered by Nemotron 3 Ultra via OpenRouter.
AI Report Quality Scan	Verified & Operational	120-second timeout with resilient error handling.

23. Future Enhancements

Direct ERP Connectors: Direct SAP S/4HANA and NetSuite procurement API sync.
Supplier Action Plans: AI-generated decarbonization recommendations for high-emitting vendors.
Category 4 Logistics Engine: Automated freight emissions calculation using tonne-kilometre models.
Cryptographic Audit Anchoring: Immutable ledger verification for third-party carbon credits.

24. Conclusion

TraceCarbon v2.10.1 delivers a robust, secure, and mathematically reconciled platform for enterprise Scope 3 greenhouse gas accounting. By combining agentic AI ingestion with a multi-tier calculation engine and zero-trust evidence security, TraceCarbon bridges the gap between high-level macroeconomic estimates and primary supplier operational reality.

25. References and Dataset Attributions

1. **GHG Protocol**: *Corporate Value Chain (Scope 3) Accounting and Reporting Standard*, World Resources Institute (WRI) & WBCSD.

2. **Central Electricity Authority (CEA)**: *CO2 Baseline Database for the Indian Power Sector*, Ministry of Power, Govt. of India, User Guide Version 19.0 (Dec 2023) and Version 20.0 (Dec 2024).

3. **IPCC**: *2006 IPCC Guidelines for National Greenhouse Gas Inventories*, Vol. 2 (Energy), Ch. 2 (Stationary Combustion), Table 2.2.

4. **Open CEDA by Watershed**: Comprehensive Environmental Data Archive, based on US BEA USEEIO v2.0.1 model, released under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0).

5. **SEBI BRSR Principle 6 Disclosures**: Audited filings of JSW Steel, Apollo Tyres, Hindalco, UltraTech, Tata Chemicals, JK Paper, Arvind, Asahi Glass, Supreme Industries, and BHEL (FY 2022–23 and FY 2023–24).

End of Complete Technical Architecture, Implementation & Analytics Report.